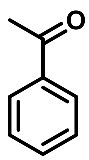
Chem 0345
Activity 9: EAS
Extra Practice Problems — Key

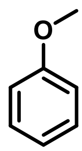
1) Draw resonance structures of the following compounds. Specify if the substituent is an electron donating group (EDG) or electron withdrawing group (EWG) with respect to the benzene ring.

A.



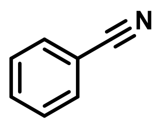
EWG

B.



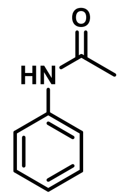
EDG

C.



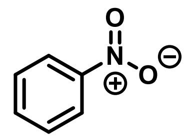
EWG

D.



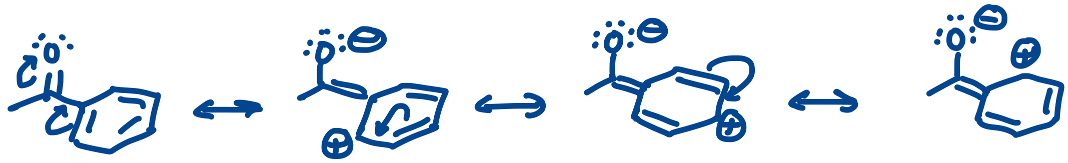
EDG

E.

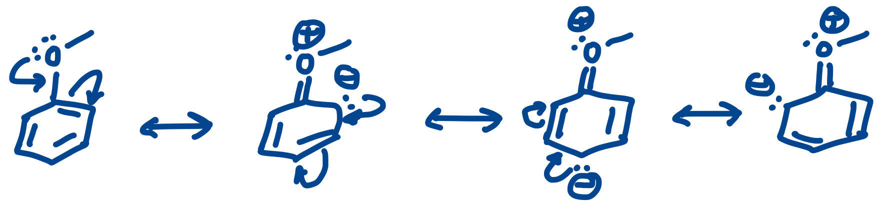


EWG

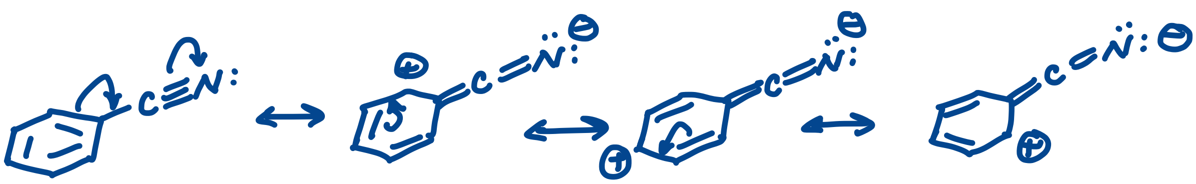
A.



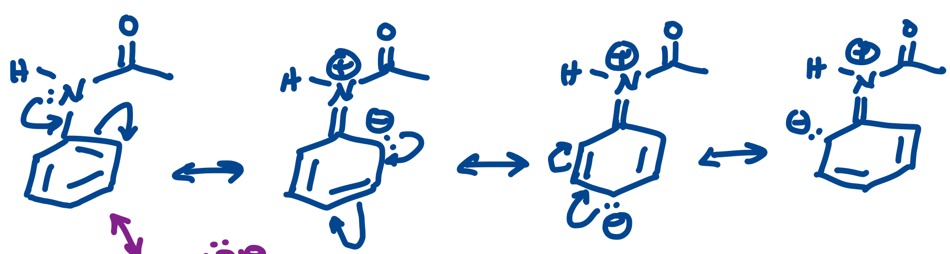
B.



C.

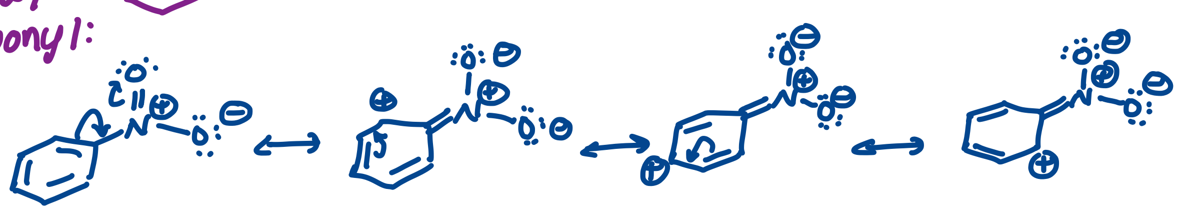


D.

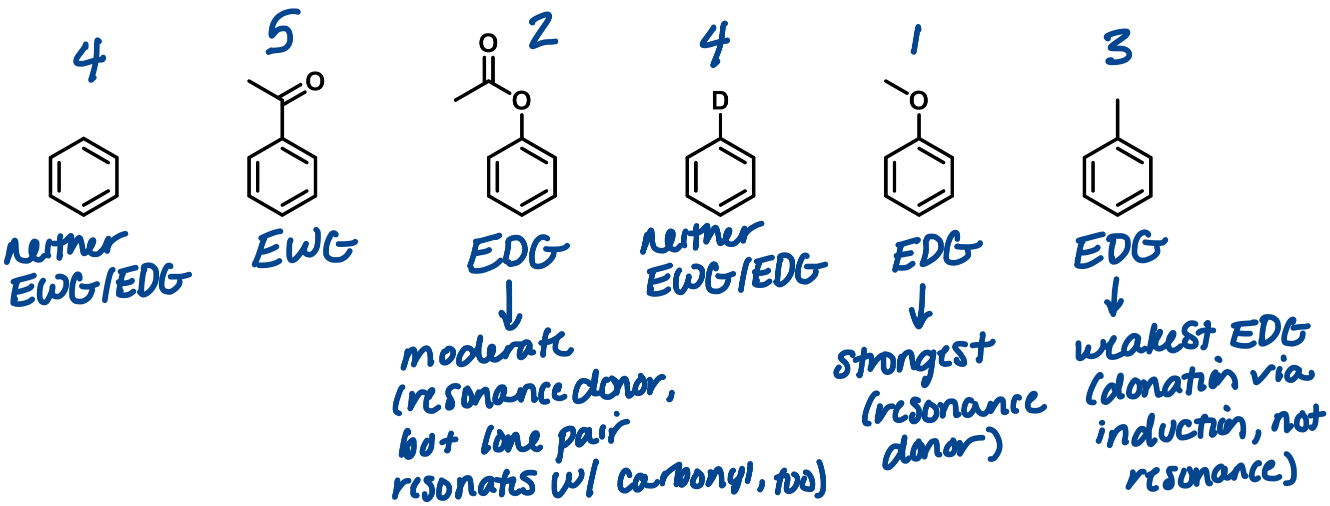


N lone pair can also resonate w/ carbonyl:

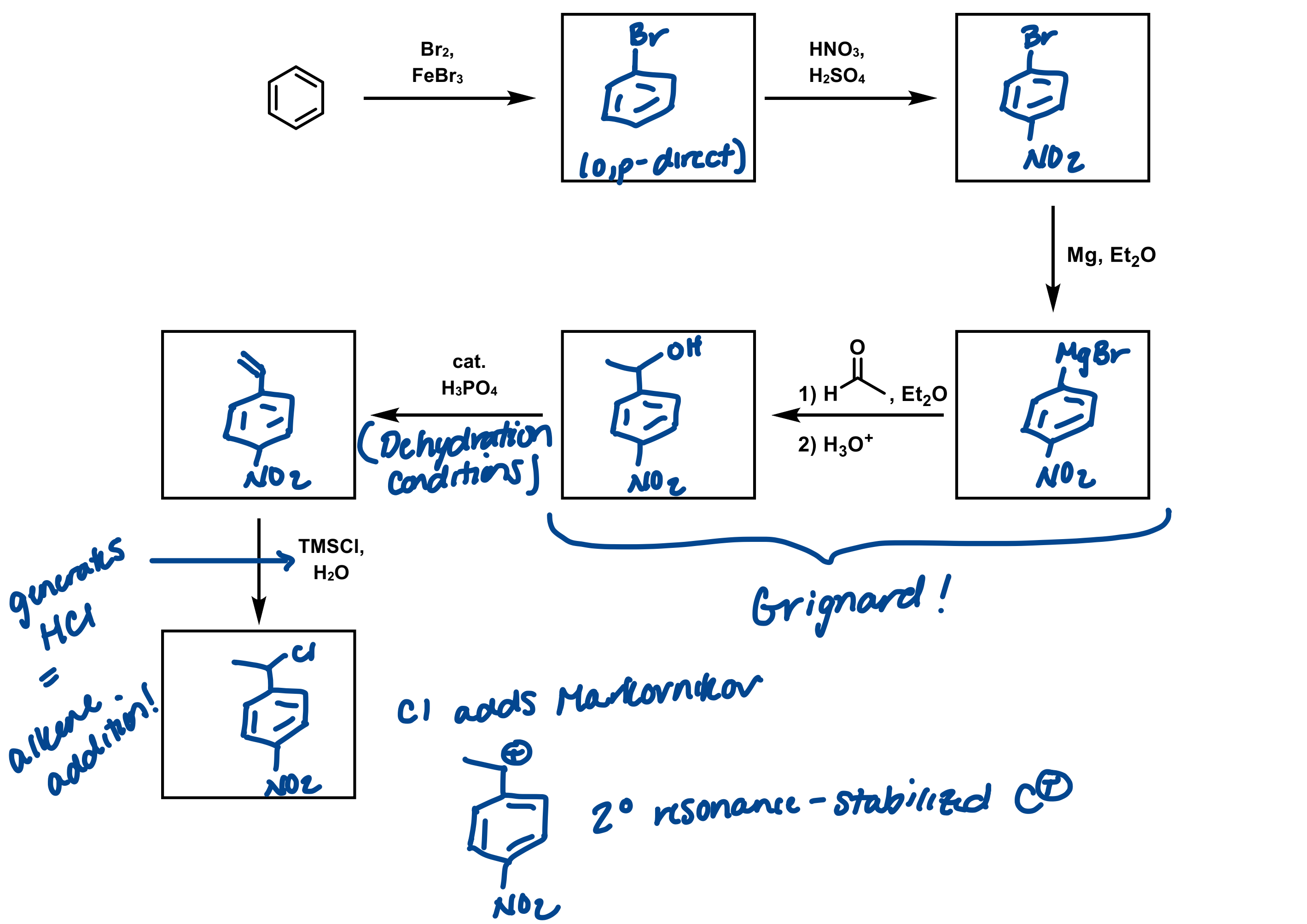
E.



2) Rank the following in following molecules in terms of reacting with Cl_2 and $AlCl_3$. 1 is the fastest, 5 is the slowest. (Two of these molecules have the same rate of reactivity.)

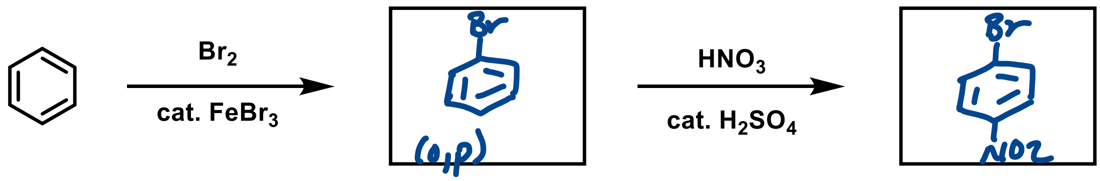


3) Fill in the boxes with the major deuterium-containing compound.

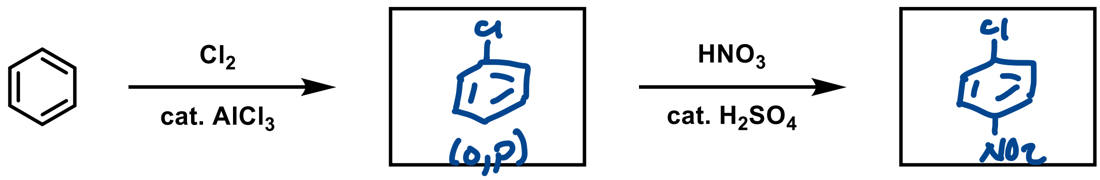


4) Predict the major product of the following reactions.

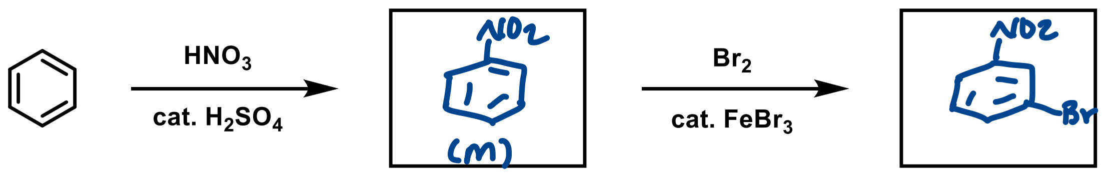
A.



B.

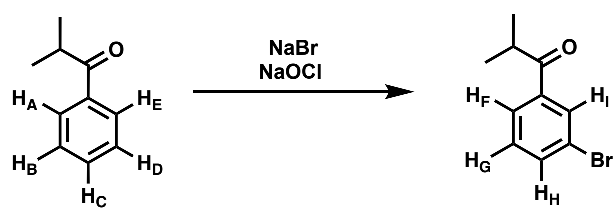


C.



D.

5) The following EAS reaction is performed and a ^{1}H NMR spectrum of the reaction mixture is obtained in CD_3OD:



A. Predict the splitting pattern of each proton in the starting material and product. Assume all 3-bond and 4-bond aromatic coupling interactions are observed in the ^{1}H NMR spectrum of the molecule shown below. Assume all $^3J_{Aromatic} > {}^4J_{Aromatic}$.

Assume all ^3J are equiv. and all ^4J are equiv

H_A = doublet of doublets (H_B) (H_C)

H_B = triplet (H_A/H_C)

H_C = triplet of triplets (H_B/H_C) (H_A/H_E)

H_D = triplet (H_C/H_E)

H_E = doublet of doublets (H_D) (H_A)

H_A and H_C are equivalent

- and -

H_B and H_D are equivalent

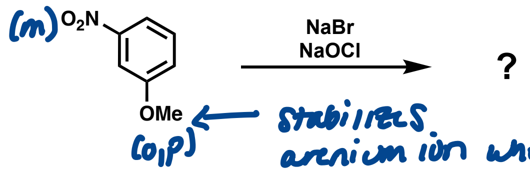
H_F = doublet of triplets (H_G) (H_H/H_I)

H_G = triplet (H_F/H_H)

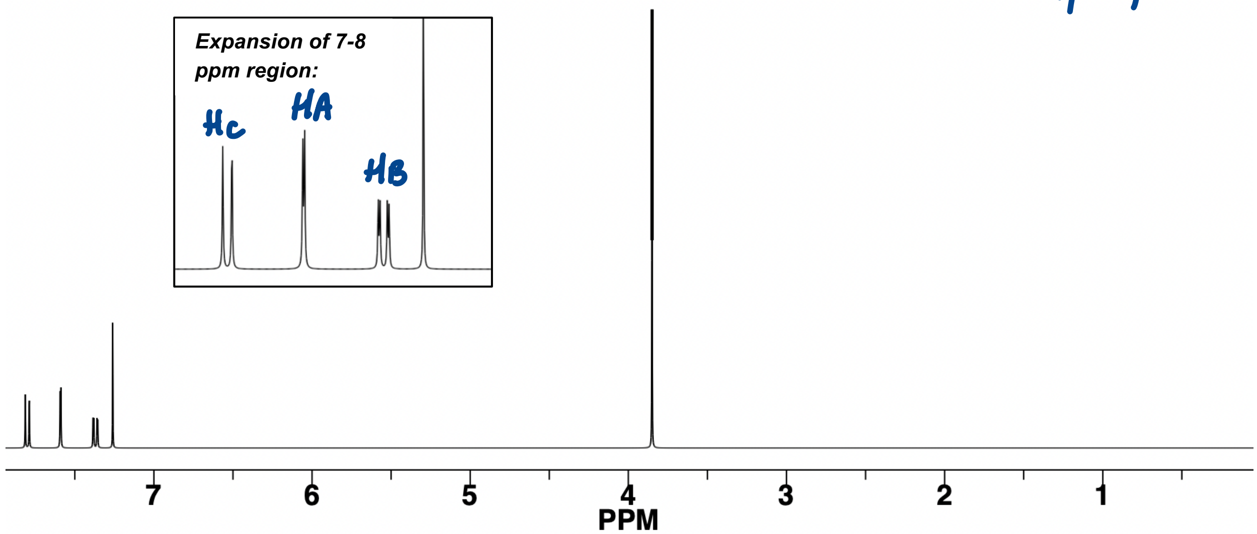
H_H = doublet of triplets (H_G) (H_F/H_I)

H_I = triplet (H_F/H_H)

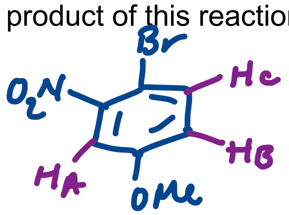
6) A ^{1}H NMR spectrum of the product of the monobromination reaction below is obtained in $CDCl_3$:



stabilizes arenium ion when electrophile adds to o,p-positions



A. Draw the product of this reaction.



B. Label each aromatic proton on your product structure. For each proton:

i) Predict the splitting pattern.

HA: doublet (through 4-bond coupling = smaller J)

HB: doublet of doublets

Hc: doublet (through 3-bond coupling = larger J)

ii) Predict if the proton's signal will be upfield or downfield relative to benzene.

HA: ortho to EDG = upfield

HB: ortho to EDG = upfield

Hc*: meta to EDG and EWG, so no predictable resonance effect ortho to inductively withdrawing Br = downfield

iii) Assign each proton to a signal in the NMR spectrum. Does your expected splitting pattern/chemical shift match the spectrum?

yup! 🎉😅

* Note: this is a tricky one and not at the level you are expected to predict for quizzes/exams